

**Example 9.16**

The initial- and final-value theorems can be useful in checking the correctness of the Laplace transform calculations for a signal. For example, consider the signal  $x(t)$  in Example 9.4. From eq. (9.24), we see that  $x(0^+) = 2$ . Also, using eq. (9.29), we find that

$$\lim_{s \rightarrow \infty} sX(s) = \lim_{s \rightarrow \infty} \frac{2s^3 + 5s^2 + 12s}{s^3 + 4s^2 + 14s + 20} = 2,$$

which is consistent with the initial-value theorem in eq. (9.110).

**9.5.11 Table of Properties**

In Table 9.1, we summarize the properties developed in this section. In Section 9.7, many of these properties are used in applying the Laplace transform to the analysis and characterization of linear time-invariant systems. As we have illustrated in several examples, the various properties of Laplace transforms and their ROCs can provide us with

**TABLE 9.1** PROPERTIES OF THE LAPLACE TRANSFORM

| Section                           | Property                                                                                                  | Signal                            | Laplace Transform                                                 | ROC                                                                      |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------|
|                                   |                                                                                                           | $x(t)$                            | $X(s)$                                                            | $R$                                                                      |
|                                   |                                                                                                           | $x_1(t)$                          | $X_1(s)$                                                          | $R_1$                                                                    |
|                                   |                                                                                                           | $x_2(t)$                          | $X_2(s)$                                                          | $R_2$                                                                    |
| 9.5.1                             | Linearity                                                                                                 | $ax_1(t) + bx_2(t)$               | $aX_1(s) + bX_2(s)$                                               | At least $R_1 \cap R_2$                                                  |
| 9.5.2                             | Time shifting                                                                                             | $x(t - t_0)$                      | $e^{-st_0}X(s)$                                                   | $R$                                                                      |
| 9.5.3                             | Shifting in the $s$ -Domain                                                                               | $e^{s_0 t}x(t)$                   | $X(s - s_0)$                                                      | Shifted version of $R$ (i.e., $s$ is in the ROC if $s - s_0$ is in $R$ ) |
| 9.5.4                             | Time scaling                                                                                              | $x(at)$                           | $\frac{1}{ a }X\left(\frac{s}{a}\right)$                          | Scaled ROC (i.e., $s$ is in the ROC if $s/a$ is in $R$ )                 |
| 9.5.5                             | Conjugation                                                                                               | $x^*(t)$                          | $X^*(s^*)$                                                        | $R$                                                                      |
| 9.5.6                             | Convolution                                                                                               | $x_1(t) * x_2(t)$                 | $X_1(s)X_2(s)$                                                    | At least $R_1 \cap R_2$                                                  |
| 9.5.7                             | Differentiation in the Time Domain                                                                        | $\frac{d}{dt}x(t)$                | $sX(s)$                                                           | At least $R$                                                             |
| 9.5.8                             | Differentiation in the $s$ -Domain                                                                        | $-tx(t)$                          | $\frac{d}{ds}X(s)$                                                | $R$                                                                      |
| 9.5.9                             | Integration in the Time Domain                                                                            | $\int_{-\infty}^t x(\tau)d(\tau)$ | $\frac{1}{s}X(s)$                                                 | At least $R \cap \{\operatorname{Re}\{s\} > 0\}$                         |
| Initial- and Final-Value Theorems |                                                                                                           |                                   |                                                                   |                                                                          |
| 9.5.10                            | If $x(t) = 0$ for $t < 0$ and $x(t)$ contains no impulses or higher-order singularities at $t = 0$ , then |                                   |                                                                   |                                                                          |
|                                   |                                                                                                           |                                   | $x(0^+) = \lim_{s \rightarrow \infty} sX(s)$                      |                                                                          |
|                                   | If $x(t) = 0$ for $t < 0$ and $x(t)$ has a finite limit as $t \rightarrow \infty$ , then                  |                                   |                                                                   |                                                                          |
|                                   |                                                                                                           |                                   | $\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} sX(s)$ |                                                                          |

considerable information about a signal and its transform that can be useful either in characterizing the signal or in checking a calculation. In Sections 9.7 and 9.8 and in some of the problems at the end of this chapter, we give several other examples of the uses of these properties.

## 9.6 SOME LAPLACE TRANSFORM PAIRS

As we indicated in Section 9.3, the inverse Laplace transform can often be easily evaluated by decomposing  $X(s)$  into a linear combination of simpler terms, the inverse transform of each of which can be recognized. Listed in Table 9.2 are a number of useful Laplace

**TABLE 9.2** LAPLACE TRANSFORMS OF ELEMENTARY FUNCTIONS

| Transform pair | Signal                                                            | Transform                                        | ROC                  |
|----------------|-------------------------------------------------------------------|--------------------------------------------------|----------------------|
| 1              | $\delta(t)$                                                       | 1                                                | All $s$              |
| 2              | $u(t)$                                                            | $\frac{1}{s}$                                    | $\Re\{s\} > 0$       |
| 3              | $-u(-t)$                                                          | $\frac{1}{s}$                                    | $\Re\{s\} < 0$       |
| 4              | $\frac{t^{n-1}}{(n-1)!}u(t)$                                      | $\frac{1}{s^n}$                                  | $\Re\{s\} > 0$       |
| 5              | $-\frac{t^{n-1}}{(n-1)!}u(-t)$                                    | $\frac{1}{s^n}$                                  | $\Re\{s\} < 0$       |
| 6              | $e^{-\alpha t}u(t)$                                               | $\frac{1}{s + \alpha}$                           | $\Re\{s\} > -\alpha$ |
| 7              | $-e^{-\alpha t}u(-t)$                                             | $\frac{1}{s + \alpha}$                           | $\Re\{s\} < -\alpha$ |
| 8              | $\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(t)$                         | $\frac{1}{(s + \alpha)^n}$                       | $\Re\{s\} > -\alpha$ |
| 9              | $-\frac{t^{n-1}}{(n-1)!}e^{-\alpha t}u(-t)$                       | $\frac{1}{(s + \alpha)^n}$                       | $\Re\{s\} < -\alpha$ |
| 10             | $\delta(t - T)$                                                   | $e^{-sT}$                                        | All $s$              |
| 11             | $[\cos \omega_0 t]u(t)$                                           | $\frac{s}{s^2 + \omega_0^2}$                     | $\Re\{s\} > 0$       |
| 12             | $[\sin \omega_0 t]u(t)$                                           | $\frac{\omega_0}{s^2 + \omega_0^2}$              | $\Re\{s\} > 0$       |
| 13             | $[e^{-\alpha t} \cos \omega_0 t]u(t)$                             | $\frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2}$ | $\Re\{s\} > -\alpha$ |
| 14             | $[e^{-\alpha t} \sin \omega_0 t]u(t)$                             | $\frac{\omega_0}{(s + \alpha)^2 + \omega_0^2}$   | $\Re\{s\} > -\alpha$ |
| 15             | $u_n(t) = \frac{d^n \delta(t)}{dt^n}$                             | $s^n$                                            | All $s$              |
| 16             | $u_{-n}(t) = \underbrace{u(t) * \cdots * u(t)}_{n \text{ times}}$ | $\frac{1}{s^n}$                                  | $\Re\{s\} > 0$       |